Notes on College Algebra

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This is a self-taught journey ever since 2024. I expect myself to finish doing College Algebra and Trigonometry by the middle of summer break. The notes I am covering right now are from College Algebra. Trigonometry is separate, but I will post one with both documents merged together. Errata is from my own accord. List of notes can be found on my website, the main page is `here`. These notes are from many books. The list will be provided in the end of this document as a `.bib` file.

These notes have hand-selected topics, and are not following any syllabus. The topics can be all over the place, and they can be very random or specific.

Contents

§1 Solving Quadratic Systems

I did not actually include how to solve quadratic equations (or whatever in a regular College Algebra textbook involving sections with quadratics yada yada) because I already knew how to do this. Weh.

Soon to add:

- Graphing quadratic equations.
- Solving radical and rational equations.

§1.1 One Linear + One Quadratic

A system of quadratic equations like the form mentioned can be of many forms, one can include

$$\begin{cases} A_1x^2 + B_1y^2 = C_1 \\ A_2x + B_2y = C_2 \end{cases}$$

or, still considered a quadratic in two unknowns:

$$\begin{cases} A_1x + xy + B_1y = C_1 \\ A_2x + B_2y = C_2 \end{cases}$$

and much more.

Method 1. Consider:

1. Substitute a variable from the linear equation to the quadratic.

$$\begin{cases} x^2 + y^2 = 25 & (1) \\ x - 7y = -25 & (2) \end{cases}$$

$$\begin{aligned} x - 7y &= -25 & (1) \\ x &= 7y - 25 & (2) \end{aligned}$$

2. Substitute $x = 7y - 25$ to (1) and solve for y.

3. Substitute y to any equation and solve for x.

§1.2 Both Equations Have Unknowns as Squares

This can be done simply by elimination.

Method 2. Consider:

1. Eliminate an unknown of choice.

$$\begin{cases} x^2 + y^2 = 34 & (1) \\ x^2 - y^2 = 16 & (2) \end{cases}$$

$$\begin{array}{rcl} x^2 + y^2 & = & 34 \\ x^2 - y^2 & = & 16 \\ \hline 2x^2 & = & 50 \end{array}$$

Solve for x:

$$\begin{aligned} x^2 &= 25 \\ x &= \pm 5 \end{aligned}$$

2. Substitute the x value for each case and solve for y.

3. Substitute the y value for each case and solve for x.

§1.3 Both Equations Are Same Format (With an xy Term)

It can be either done by:

Method 3 (Elimination)**.** Consider:

1. Multiply such that the constant is zero.

$$\begin{cases} 2x^2 + 5xy - 10y^2 = 8 & (1) \\ x^2 - 2xy + 3y^2 = 3 & (2) \end{cases}$$

$$\begin{array}{rl} -6x^2 - 15xy + 30y^2 & = -24 \\ 8x^2 - 16xy + 24y^2 & = 24 \\ \hline 2x^2 - 31xy + 54y^2 & = 0 \end{array}$$

2. Solve for x or y, of your desired method.

$$2x^2 - 31xy + 54y^2 = 0 \quad (3)$$

$$(x - 2y)(2x - 27y) = 0 \quad (4)$$

$$x = 2y, \frac{27}{2}y \quad (5)$$

3. Substitute each case to which equation seems more of ease to deal.

Method 4 (By Substituting $y = tx$)**.** Consider:

1. Substitute $y = tx$.

$$\begin{cases} 2x^2 + 5xy - 10y^2 = 8 & (1) \\ x^2 - 2xy + 3y^2 = 3 & (2) \end{cases}$$

   becomes

$$\begin{cases} 2x^2 + 5tx^2 - 10t^2x^2 = 8 & (3) \\ x^2 - 2tx^2 + 3t^2x^2 = 3 & (4) \end{cases}$$

2. Solve for x^2, not x[1]. Be not mistaken. Factor out x^2 first then divide both parts by the factored form. Steps are skipped for, unfortunately, simplicity reasons.

$$x^2 = \frac{8}{2 + 5t - 10t^2} \text{ for } (3) \tag{5}$$

$$x^2 = \frac{3}{1 - 2t + 3t^2} \text{ for } (4) \tag{6}$$

3. Equate both terms; solve for t.

4. Substitute each case of t to either equation from the simplified forms.

§1.4 Symmetric Equations in Quadratic Systems

An equation is considered *symmetric* (for two unknowns) if interchanging any unknown won't change the value of the equation.

$$2x^2 - xy + 2y^2 - x - y$$

is considered symmetric because:

$$\begin{aligned} 2x^2 + 2y^2 &\quad \text{is symmetric.} \\ -xy &\quad \text{is symmetric.} \\ -x - y &\quad \text{is symmetric.} \end{aligned}$$

$-x - y$ is symmetric because we can factor the minus sign to get $-(x + y)$. $x + y$ is ALWAYS symmetric.

[1] You can solve for x but I'd rather just leave as is, but it would work the same if you solved for x separately.

We can solve a system of quadratic equations that have two symmetric quadratics by two terms by replacing one unknown with some arbitrary values p and q, with $p+q$ and another with $p-q$.

Method 5. Consider:

1. Substitute an unknown with above. We let $x = p+q$ and $y = p-q$.

$$\begin{cases} 3x^2 - 2xy + 3y^2 + x + y - 50 = 0 & (6) \\ 2x^2 + 2y^2 - 3x - 3y - 29 = 0 & (7) \end{cases}$$

   becomes, after factoring the first equation and dividing the factor:

$$\begin{cases} 2p^2 + 4q^2 + p - 25 = 0 & (3) \\ 4p^2 + 4q^2 - 6p - 29 = 0 & (4) \end{cases}$$

2. Subtract (3) from (4) or vice versa and multiply everything by negative one if the other way.

3. Solve the newly eliminated quadratic by any method.

$$(4p^2 + 4q^2 - 6p - 29) - (2p^2 + 4q^2 + p - 25) = 0 \tag{8}$$
$$2p^2 - 7p - 4 = (p-4)(2p+1) = 0 \tag{9}$$
$$p = \boxed{4, -\frac{1}{2}} \tag{10}$$

4. Substitute each case one-by-one to any equation. We will choose (3). Solve for q:

   a) For $p = 4$, $q = \pm\frac{\sqrt{11}}{2}i$.

   b) For $p = -\frac{1}{2}$, $q = \pm\frac{5}{2}$

5. Re-substitute each value for $x = p+q$ and $y = p-q$.

   a) For $p = 4$, $q = \pm\frac{\sqrt{11}}{2}i$, there will be two cases each unknown:

$$x = 4 \pm \frac{\sqrt{11}}{2}i$$
$$y = 4 \mp \frac{\sqrt{11}}{2}i$$

   b) For $p = -\frac{1}{2}$, $q = \pm\frac{5}{2}$, there will be two cases each unknown:

$$x = 2, y = -3$$
$$x = -3, y = 2$$

$$\therefore \boxed{\left\{\left(4+\frac{\sqrt{11}}{2}i, 4-\frac{\sqrt{11}}{2}i\right), \left(4-\frac{\sqrt{11}}{2}i, 4+\frac{\sqrt{11}}{2}i\right), (2,-3), (-3,2)\right\}}$$

§1.5 A Special Kind of Symmetric System

If a system is in the form,

$$\begin{cases} x^2+y^2=C_1 \\ A_1B_1xy=C_2 \end{cases}$$

There is a more simpler method to solve this. However, not all systems in this form are easily solvable like this. But it is worth a chance if you pray that the system is like this by quickly inspecting.

Method 6. Consider:

$$\begin{cases} x^2+y^2=25 \\ xy=12 \end{cases}$$

1. Multiply the second equation by 2 to get $2xy = 24$.

2. Add the first equation to it.

$$\begin{aligned} x^2+2xy+y^2 &= 25-24 \\ (x+y)^2 &= 49 \\ x+y &= \pm 7 \end{aligned}$$

3. Do the same, but this time subtract.

$$\begin{aligned} x^2-2xy+y^2 &= 25-24 \\ (x-y)^2 &= 1 \\ x-y &= \pm 1 \end{aligned}$$

4. Form a set of linear systems from each case; each system's solution is a solution of the original system.

$$\begin{cases} x+y=7 \\ x-y=1 \end{cases} \quad \begin{cases} x+y=7 \\ x-y=-1 \end{cases} \quad \begin{cases} x+y=-7 \\ x-y=1 \end{cases} \quad \begin{cases} x+y=-7 \\ x-y=-1 \end{cases}$$

5. Solve for each system. Hence,

$$\therefore \boxed{\{(4,3),(3,4),(-3,-4),(-4,-3)\}}$$

§1.6 As a System with a Cubic + Linear

I will make this brief. In the form of,

$$\begin{cases} A_1^3 + B_1^3 = C_1 \\ A_1x + B_1y = C_2 \end{cases}$$

Despite it has a cubic, we can see it is a sum of cubes. If ever, the second equation is divisible by the factored form of the first equation, divide it by the second equation.

$$\begin{cases} (A_1x + B_1y)(A_1^2x^2 + A_1B_1xy + B_1^2y^2) \\ A_1x + B_1y = C_2 \end{cases}$$

$$\frac{\cancel{(A_1x + B_1y)}(A_1^2x^2 + A_1B_1xy + B_1^2y^2)}{\cancel{A_1x + B_1y}} = (A_1^2x^2 + A_1B_1xy + B_1^2y^2)$$

Solve $A_1x + B_1y = C_2$ for y.

$$y = \frac{C_1}{B_1} - \frac{A_1}{B_1}x$$

To continue, substitute this value of y to $A_1^2x^2 + A_1B_1xy + B_1^2y^2$. Then, re-substitute x from the simplified form to the original y.

§1.7 With a Homogeneous Equation

A *homogeneous equation* is an equation that has a trivial solution of 0. It must satisfy one of these conditions. More formally, each term has the same degree.

$$3x + 2y = 0$$

is a homogeneous equation because it has a trivial solution of 0.

$$2x^2 - 3xy + y^2 = 0$$

is a homogeneous equation because all terms are the same degree of two.

$$2x + 4y - 2 = 0$$

is not a homogeneous equation because it does not have all terms on the same degree and has no trivial solution. We can redefine a homogeneous equation to not have a constant term, because immediately it won't not have a trivial solution of 0.

With a homogeneous quadratic,

$$\begin{cases} 6x^2 - 7xy + 2y^2 = 0 \\ y = x^2 - 4 \end{cases}$$

The first equation is considered homogeneous. Start solving by: 1. Solve by letting $y = tx$. Substitute this to the first equation.

$$6x - 7tx^2 + 2t^2x^2$$

2. Divide x^2. ___Do not___ let $x^2 = 0$. An extraneous solution $x = 0, y = 0$ will concur, and is not ALWAYS a solution and is not proper in the steps mentioned.

$$\begin{aligned} 2t^2 - 7t + 6 &= 0 \\ t &= 2, \frac{3}{2} \end{aligned}$$

3. Re-substitute t to $y = tx$ for each case to the second equation from the solutions of t. Then re-substitute these forms of y to the second equation of the system.

$$\begin{aligned} y = 2x \rightarrow 2x &= x^2 - 4 \\ x^2 - 2x - 4 &= 0 \\ x &= \boxed{1 \pm \sqrt{5}} \\ y = \frac{3}{2}x \rightarrow \frac{3}{2}x &= x^2 - 4 \\ 2x^2 - 3x - 8 &= 0 \\ x &= \boxed{\frac{3}{4} \pm \frac{\sqrt{73}}{4}} \end{aligned}$$

4. Re-substitute both solutions to their respective values of y.

For $y = 2x$, $x = 1 \pm \sqrt{5}$:

$$y = 2(1 \pm \sqrt{5}) = 2 \pm 2\sqrt{5}$$

For $y = \frac{3}{2}x$, $\frac{3}{4} \pm \frac{\sqrt{73}}{4}$:

$$y = \frac{3}{2}\left(\frac{3}{4} \pm \frac{\sqrt{73}}{4}\right) = \frac{9}{8} \pm \frac{3\sqrt{73}}{8}$$

5. Construct the solution set.

$$\therefore \left\{(1 + \sqrt{5}, 2 + 2\sqrt{5}), (1 - \sqrt{5}, 2 - 2\sqrt{5})\right\}$$

$$\wedge \left\{\left(\frac{3}{4} + \frac{\sqrt{73}}{4}, \frac{9}{8} + \frac{3\sqrt{73}}{8}\right), \left(\frac{3}{4} - \frac{\sqrt{73}}{4}, \frac{9}{8} - \frac{3\sqrt{73}}{8}\right)\right\}$$

There are four solutions, just split between sets.

§2 Inequalities

An inequality is considered *absolute* if every solution applies to the inequality.

$$x^2 + 1 > 0 \text{ applies for } x \in \mathbb{R}$$

An inequality is *conditional* if only certain solutions apply to the inequality.

$$x^2 + 2x + 1 > 0 \text{ applies only } x > -1$$

§2.1 Certain Properties of Inequalities

We shall bring upon our first theorems that apply:

Theorem 2.1

Our first theorems apply:

1. $a < b \wedge b < c \Rightarrow a < c$
2. $a < b \Rightarrow a + c < b + c$
3. $a < b \wedge c > 0 \Rightarrow ac < bc$
4. $a < b \wedge c < 0 \Rightarrow ac > bc$